

SPECTRAVIDEO USERS GROUP -- Wgtn

Newsletter Number 17 -- December 1985

The next meeting is on Monday, February 17th at 7.30 in the Staff Common Room, Wellington Clinical School, Mein Street Newtown.

At February's meeting Darryl's tutorial on data files will take place as it was abandoned last month.

There will be no meeting in December or in January.

Wellington SV Users Group

Comittee Members

Chairman ----- Darryl Alison Ph 85522 (Paraparaumu)

Secretary ----- Joseph Sol

Treasurer ----- Ross Fletcher 788454

Garry Clark 367872

Newsletter Committee

Steve Hanson 729459 (work)

John Matthews 327805

Other specific areas of responsibility for committee members are

Garry -- Radio Access & public domain tape software;

John -- Library;

Darryl -- Program evaluation committee chairman & Kapiti users representative.

Club Subscriptions

The club subscription is now \$15.00/year.

New members this month include Philip Daymond and Paul Lawrence.

Note that expiry date for subscriptions is printed on the address label of newsletter. Subscriptions not renewed a month after this date will not receive newsletters from then.

Newsletter Contributions

If you are submitting programs to go in the newsletter they MUST satisfy the following conditions or they will NOT be printed.

1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page and an inch at least around the sides

2 ... typed with a dark ribbon OR printed using emphasized print mode

3 ... include a description of what the program does , typed on A4 paper with example input

4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette

5 ... line numbers should be continuous -- do a RENUM before printing the programs

6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .

7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy I know .

All other newsletter contributions should be sent to The Editor, P.O Box 26050, Newlands. They may miss the next newsletter if sent anywhere else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30th/31st of the month previous to publication.

Please follow these guidelines, articles with a 4mm edge all around can not be printed , and neither can articles produced on a dot matrix printer with low resolution . They are simply unreadable after photocopying.

Basic 80column Patch

```
2 REM Disk Basic, like the distributed CP/M comes to us in NTSC
4 REM format. This patches Disk Basic to stop the screen wobble
6 REM NTSC causes on PAL televisions/monitors.
10 CLEAR 500:FIELD #0, 128 AS A$, 128 AS B$
20 D$=DSKI$(1,0,3)
30 FOR X=0 TO 7:READ Z$:POKE VARPTR(#0)+126+X,VAL("&H"+Z$):NEXT
40 DSKO$ 1,0,3
50 DATA 6C, 50, 58, 0B, 26, 02, 18, 1E
```

Palmerston North Subgroup

The Palmerston North subgroup are holding meetings on the 1st Monday of each month at THE COMPUTER EXPERIENCE in Palmerston North.

Club Sponsor

MicroStyle Computers -- 29 High Street, Lower Hutt

Photocopying from Library

Anybody requiring articles from the library to be photocopied please send request to the LIBRARIAN, not any other committee member. We have had problems with articles not being copied when requested, please ensure that requests go to the librarian.

Also membership renewals should be sent to Box 26050, Newlands, Wellington, not to any other box, otherwise it is possible that the mailing list will not get updated.

The following program, written by Darryl, is a filer for recipes. It saves data to tape, or disc if you enter drive number as well as filename (ie 1:recipe). This is not a fancy program, but it does provide a mini-database function that you can expand on. Below is an example of its printer output. Battenberg cake

Suitable for freezing but add almond paste after thawing. Serves 10

8oz Butter
8oz Caster sugar
4 Eggs
8oz Self-raising flour
4 tbsp Milk
Red food colouring
3 tbsp Apricot jam
8oz Almond paste

Preheat the oven to 375oF (190oC), Gas 5. Grease and line swiss roll tin, Make pleat in paper down centre to divide in half lengthways. Cream butter and sugar. Beat in eggs. Sift flour and fold into creamed mixture with milk. Spread half mixture into one side of tin. Add a few drops of food colouring to remaining mixture and spread into other half of tin. Bake for about 45 minutes. Cut each cake in half lengthways. Warm jam and use to stick cake pieces together in chequerboard pattern. Spread jam over cake. Sprinkle caster sugar over working surface roll out almond paste and wrap round cake. Crimp edges and make diamond pattern on top using a sharp knife

```

CLS: CLEAR50: CLEARFRE(0) - 6000: WIDTH40: SCREEN, 0: AR=30: L=7: SP$=STRING$(39, 32): DI
$(AR, AR), WU$(AR, L), G$(AR), NM$(AR): P$=CHR$(27)+"p": Q$=CHR$(27)+"q"
ONSTOPGOSUB70: STOPON
CLS: LOCATE,, 0: PRINTSPC(11): P$= MENU SERVICE "Q$
LOCATE11, 3: PRINT"1 = SAVE RECIPE": LOCATE11, 5: PRINT"2 = LOAD RECIPE": LOCATE11,
PRINT"3 = PRINT RECIPE": LOCATE11, 9: PRINT"4 = EDIT RECIPE": LOCATE11, 11: PRINT"5
NEW RECIPE": LOCATE11, 13: PRINT"6 = EXIT COOKIE"
A$=INKEY$: IFA$<"1"ORA$>"6"THEN50
ONVAL (A$)GOSUB80, 100, 580, 350, 120, 70: GOTO50
WIDTH39: CLEAR200: SCREEN, 1: LOCATE,, 1: END
CLS: PRINTSPC(17): P$= SAVE "Q$: LOCATE0, 3: PRINT"Ready to save your RECIPES to t
e Y/N": GOSUB340: IFA$="N"ORA$="n"THENRETURN30
PRINT: INPUT"Filename": N$: OPENN$FOROUTPUTAS#1: FORI=0TOAR: PRINT#1, G$(I): PRINT#1
M$(I): FORJ=0TOL: PRINT#1, WU$(I, L): NEXT: FORJ=0TOAR: PRINT#1, I$(I, J): NEXTJ, I: CLOSE
RETURN30
CLS: PRINTSPC(17): P$= LOAD "Q$: LOCATE0, 3: PRINT"Ready to LOAD your RECIPES fro
tape Y/N": GOSUB340: IFA$="N"ORA$="n"THENRETURN30
PRINT: INPUT"Filename": N$: OPENN$FORINPUTAS#1: FORI=0TOAR: LINEINPUT#1, G$(I): LIN
PUT#1, NM$(I): FORJ=0TOL: LINEINPUT#1, WU$(I, L): NEXT: FORJ=0TOAR: LINEINPUT#1, I$(I,
NEXTJ, I: CLOSE#1: RETURN30
CLS: PRINTSPC(11): P$= NEW RECIPE "Q$: FORI=0TOAR: IFNM$(I)=""THEN140ELSENEXT
PRINT: PRINT"There is no spare room in this file for more recipes please star
another file": FORZ=1TO3000: NEXT: RETURN30
LOCATE0, 3, 1: PRINT"Please enter the title for recipe no. "I"?": LINEINPUTNM$
LOCATE0, 6: PRINT"Please give a general write up about therecipe e.g how many
ple it will feedor any special things about it"
VPOKE615, 219: LOCATE0, 9
A$=INKEY$: IFVPEEK(615)<>219ORA$=CHR$(13)THEN180ELSEIFA$=CHR$(27)THENRETURN30
PRINTA$: GOTO170
LOCATE0, 17, 0: GOSUB330: IFA$="N"ORA$="n"THENLOCATE0, 18, 1: PRINTSP$: GOTO160
LOCATE5, 18: PRINT"PRESS ENTER TO CONTINUE": LOCATE0, 9: LINEINPUTG$(I)
IFA=7THENA=0: RETURN
CLS: PRINTSPC(12): P$= INGREDIENTS "Q$K: LOCATE0, 3: PRINT"Please enter the ingre
ents plus their quantities. When all ingredients have been entered type 'END
to carry on": FORL=0TOAR
LOCATE0, 8, 1: PRINTL"?": LINEINPUTI$(I, L): IFI$(I, L)="END"THENI$(I, L)="" : GOTO25
LOCATE0, 9, 0: GOSUB330: IFA$="N"ORA$="n"THENLOCATE0, 10: PRINTSP$: GOTO220
LOCATE0, 8, 1: PRINTSP$: SP$: SP$: NEXT
CLS: PRINTSPC(13): P$= INSTRUCTIONS "Q$: PRINT
PRINT"Please enter the instructions for the recipe. There is a max of eigh
blanks that can be used": PRINT: L=-1
VPOKE535, 219: LOCATE0, 7, 1
A$=INKEY$: IFVPEEK(535)<>219ORA$=CHR$(13)THEN290ELSEIFA$=CHR$(27)THENRETURN30
PRINTA$: GOTO280
LOCATE0, 14, 0: GOSUB330: IFA$="N"ORA$="n"THENLOCATE0, 15: PRINTSP$: LOCATE0, 7, 1: GO
280
L=L+1: LOCATE5, 15: PRINT"PRESS ENTER TO CONTINUE": LOCATE0, 7: LINEINPUTWU$(I, L):
RR=280TO535: VPOKER, 0: NEXT
IFA=7THENLOCATE0, 15: PRINTSP$: RETURN
LOCATE0, 6: PRINTSP$: LOCATE0, 15: PRINTSP$: IFL>6THENRETURN30ELSE270
PRINT: PRINTSPC(7)"IS THIS CORRECT (Y/N)"
A$=INKEY$: IFA$<"Y"ANDA$<"Y"ANDA$<"N"ANDA$<"n"THEN340ELSERETURN
CLS: PRINTSPC(10): P$= EDIT RECIPE "Q$: PRINT: PRINT
FORI=0TOAR
PRINTUSING"###": I$: IFNM$(I)=""THENPRINTSTRING$(30, 32)ELSEPRINTNM$(I)
IFPEEK(&HFA03)>19THEN390ELSENEXT: GOTO360

```



```

10 CLS: CLEAR50: CLEARFRE(0) - 6000: WIDTH40: SCREEN, 0: AR=30: L=7: SP$=STRING$(39,32): DI
MI$(AR,AR), WU$(AR,L), G$(AR), NM$(AR): P$=CHR$(27)+"p": Q$=CHR$(27)+"q"
20 ONSTOPGOSUB70: STOPON
30 CLS: LOCATE, 0: PRINTSPC(11); P$ " MENU SERVICE "Q$
40 LOCATE11,3: PRINT"1 = SAVE RECIPE": LOCATE11,5: PRINT"2 = LOAD RECIPE": LOCATE11,
7: PRINT"3 = PRINT RECIPE": LOCATE11,9: PRINT"4 = EDIT RECIPE": LOCATE11,11: PRINT"5
= NEW RECIPE": LOCATE11,13: PRINT"6 = EXIT COOKIE"
50 A$=INKEY$: IFA$<"1"OR A$>"6" THEN50
60 ONVAL(A$)GOSUB80,100,580,350,120,70: GOTO50
70 WIDTH39: CLEAR200: SCREEN, 1: LOCATE,,1: END
80 CLS: PRINTSPC(17); P$ " SAVE "Q$: LOCATE0,3: PRINT"Ready to save your RECIPES to t
ape Y/N": GOSUB340: IFA$="N"OR A$="n" THENRETURN30
90 PRINT: INPUT"Filename"; N$: OPENN$FOROUTPUTAS#1: FORI=0TOAR: PRINT#1, G$(I): PRINT#1
.NM$(I): FORJ=0TOL: PRINT#1, WU$(I,L): NEXT: FORJ=0TOAR: PRINT#1, I$(I,J): NEXTJ, I: CLOSE
#1: RETURN30
100 CLS: PRINTSPC(17); P$ " LOAD "Q$: LOCATE0,3: PRINT"Ready to LOAD your RECIPES fro
m tape Y/N": GOSUB340: IFA$="N"OR A$="n" THENRETURN30
110 PRINT: INPUT"Filename"; N$: OPENN$FORINPUTAS#1: FORI=0TOAR: LINEINPUT#1, G$(I): LIN
EINPUT#1, NM$(I): FORJ=0TOL: LINEINPUT#1, WU$(I,L): NEXT: FORJ=0TOAR: LINEINPUT#1, I$(I,
J): NEXTJ, I: CLOSE#1: RETURN30
120 CLS: PRINTSPC(11); P$ " NEW RECIPE "Q$: FORI=0TOAR: IFNM$(I)="" THEN140ELSENEXT
130 PRINT: PRINT"There is no spare room in this file for more recipes please star
t another file": FORZ=1TO3000: NEXT: RETURN30
140 LOCATE0,3,1: PRINT"Please enter the title for recipe no. "I"?": LINEINPUTNM$(
I)
150 LOCATE0,6: PRINT"Please give a general write up about therecipe e.g how many
people it will feedor any special things about it"
160 VPOKE615,219: LOCATE0,9
170 A$=INKEY$: IFVPEEK(615)<>219OR A$=CHR$(13) THEN180ELSE IFA$=CHR$(27) THENRETURN30
ELSEPRINTA$; GOTO170
180 LOCATE0,17,0: GOSUB330: IFA$="N"OR A$="n" THENLOCATE0,18,1: PRINTSP$: GOTO160
190 LOCATE5,18: PRINT"PRESS ENTER TO CONTINUE": LOCATE0,9: LINEINPUTG$(I)
200 IFA=7THENA=0: RETURN
210 CLS: PRINTSPC(12); P$ " INGREDIENTS "Q$: LOCATE0,3: PRINT"Please enter the ingre
dients plus their quantities. When all ingredients have been entered type 'END
' to carry on": FORL=0TOAR
220 LOCATE0,8,1: PRINTL"?": LINEINPUTI$(I,L): IFI$(I,L)="END" THENI$(I,L)="" : GOTO25
0
230 LOCATE0,9,0: GOSUB330: IFA$="N"OR A$="n" THENLOCATE0,10: PRINTSP$: GOTO220
240 LOCATE0,8,1: PRINTSP$: SP$: SP$: NEXT
250 CLS: PRINTSPC(13); P$ " INSTRUCTIONS "Q$: PRINT
260 PRINT"Please enter the instructions for the recipe. There is a max of eigh
t blanks that can be used": PRINT: L=-1
270 VPOKE535,219: LOCATE0,7,1
280 A$=INKEY$: IFVPEEK(535)<>219OR A$=CHR$(13) THEN290ELSE IFA$=CHR$(27) THENRETURN30
ELSEPRINTA$; GOTO280
290 LOCATE0,14,0: GOSUB330: IFA$="N"OR A$="n" THENLOCATE0,15: PRINTSP$: LOCATE0,7,1: GO
TO280
300 L=L+1: LOCATE5,15: PRINT"PRESS ENTER TO CONTINUE": LOCATE0,7: LINEINPUTWU$(I,L):
FORR=280TO535: VPOKE R,0: NEXT
310 IFA=7THENLOCATE0,15: PRINTSP$: RETURN
320 LOCATE0,6: PRINTSP$: LOCATE0,15: PRINTSP$: IFL>6 THENRETURN30ELSE270
330 PRINT: PRINTSPC(7)"IS THIS CORRECT (Y/N)"
340 A$=INKEY$: IFA$<"Y"AND A$<"Y"AND A$<"N"AND A$<"n" THEN340ELSERETURN
350 CLS: PRINTSPC(10); P$ " EDIT RECIPE "Q$: PRINT: PRINT
360 FORI=0TOAR
370 PRINTUSING"###"; I; : IFNM$(I)="" THENPRINTSTRING$(30,32) ELSEPRINTNM$(I)
380 IFPEEK(&HFA03)>19 THEN390ELSENEXT: GOTO360

```

```

390 PRINT:PRINT"Press any key to select recipe to edit or ESC to return to menu
or space bar to continue"
400 A$=INKEY$: IFA$="" THEN400ELSE IFA$=CHR$(27) THEN RETURN30ELSE IFA$="" THEN LOCATE0
,3:GOTO370
410 LOCATE0,20:PRINTSP$;SP$;SP$:LOCATE0,20:INPUT"Recipe No.";I:LOCATE0,21:GOSUB3
30: IFA$="n" OR A$="N" THEN350
420 CLS:PRINTSPC(10);P$ EDITING "Q$
430 LOCATE0,3:PRINT"Recipe" I":":LOCATE12,3:PRINTNM$(I):LOCATE12,3,1:LINEINPUTNM
$(I):LOCATE,,0
440 GOSUB330: IFA$="N" OR A$="n" THEN420
450 LOCATE0,5:PRINTSP$:LOCATE0,6:PRINT"General Write up":LOCATE0,9,1:PRINTG$(I):
A=7:GOSUB160
460 CLS:PRINTSPC(15);P$ EDITING "Q$:PRINT:PRINT
470 FORL=0TOAR
480 PRINTUSING"##";I: IF I$(I,L)="" THENPRINTSTRING$(30,32) ELSEPRINTI$(I,L)
490 IFPEEK(&HFA03)>19 THEN500 ELSENEXT:GOTO470
500 PRINT:PRINT"Press any key to select ingredient to edit or space bar to con
tinue selection or ESC to continue"
510 A$=INKEY$: IFA$="" THEN510ELSE IFA$=CHR$(27) THEN560ELSE IFA$="" THENLOCATE0,3:GO
T0480
520 LOCATE0,20:PRINTSP$;SP$;SP$:LOCATE0,20:INPUT"Enter no of ingredient";L: IFL<0
OR L>AR THEN520 ELSELOCATE0,21:GOSUB330: IFA$="N" OR A$="n" THEN520
530 LOCATE0,20:PRINTSP$;SP$;SP$
540 LOCATE0,20:PRINT"Ingredient" L":":LOCATE14,20:PRINTI$(I,L):LOCATE14,20,1:LIN
EINPUTI$(I,L):LOCATE0,21,0:GOSUB330: IFA$="N" OR A$="n" THEN530
550 LOCATE0,20:PRINTSP$;SP$;SP$:LOCATE0,20:PRINT"Any more alterations (Y/N)":GOS
UB340: IFA$="Y" OR A$="y" THEN460
560 CLS:PRINTSPC(15);P$ EDITING "Q$:L=0
570 LOCATE0,7:PRINTWU$(I,L):A=7:GOSUB270: IFL>6 THENRETURN30: ELSE570
580 CLS:PRINTSPC(13);P$ LISTING "Q$:PRINT:PRINT
590 FORI=0TOAR
600 PRINTUSING"##";I: IFNM$(I)="" THENPRINTSTRING$(20,32) ELSEPRINTNM$(I)
610 IFPEEK(&HFA03)>19 THEN620 ELSENEXT:GOTO590
620 PRINT:PRINT"Press any key to select recipe to view or ESC to return to menu
or space bar to continue scrolling."
630 A$=INKEY$: IFA$="" THEN630ELSE IFA$=CHR$(27) THENRETURN30ELSE IFA$="" THENLOCATE0
,3:GOTO600
640 LOCATE0,20:PRINTSP$;SP$;SP$:LOCATE0,20:INPUT"Recipe No.";I:LOCATE0,21:GOSUB3
30: IFA$="n" OR A$="N" THEN580
650 LOCATE0,20:PRINTSP$;SP$;SP$:LOCATE0,20:PRINT"Printer (Y/N)":GOSUB340: IFA$="y
" OR A$="Y" THENA=7
660 CLS:PRINTSPC(3);"Recipe";NM$(I):PRINT:PRINT: IFA=7 THENLPRINTSPC(7);"Recipe:";
NM$(I):LPRINT:LPRINT
670 PRINT" General Note":PRINTG$(I): IFA=7 THENLPRINT"General Note":LPRINTG$(I)
:LPRINT:LPRINT
680 PRINT:PRINT"PRESS ANY KEY TO CONTINUE"
690 IFINKEY$="" THEN690
700 CLS:PRINT"Ingredients":FORL=0TOAR: IF I$(I,L)="" THEN710 ELSEPRINTI$(I,L):NEXT
710 IFA=7 THENLPRINT"Ingredients":FORL=0TOAR: IF I$(I,L)="" THEN720 ELSEPRINTI$(I,L)
:NEXT
720 PRINT:PRINT"PRESS ANY KEY TO CONTINUE"
730 IFINKEY$="" THEN730
740 CLS:FORL=0TO7:PRINT"Instructions":IFWU$(I,L)="" THEN770 ELSEPRINTWU$(I,L)
750 PRINT:PRINT"PRESS ANY KEY TO CONTINUE"
760 IFINKEY$="" THEN760 ELSENEXT
770 IFA=7 THENLPRINT:LPRINT:LPRINT"Instructions":FORL=0TO7:LPRINTWU$(I,L):NEXT
780 RETURN30

```

```

390 PRINT:PRINT"Press any key to select recipe to edit or ESC to return to menu
or space bar to continue"
400 A$=INKEY$: IFA$="" THEN 400 ELSE IFA$=CHR$(27) THEN RETURN 30 ELSE IFA$="" THEN LOCATE 0,3:GOTO 370
410 LOCATE 0,20:PRINT SP$;SP$;SP$:LOCATE 0,20:INPUT"Recipe No.":I:LOCATE 0,21:GOSUB 330
420 IFA$="n" OR A$="N" THEN 350
430 CLS:PRINT SP$(10);P$ EDITING "Q$
440 LOCATE 0,3:PRINT"Recipe "I":;:LOCATE 12,3:PRINT NM$(I):LOCATE 12,3,1:LINE INPUT NM$(I):LOCATE,,0
450 GOSUB 330: IFA$="N" OR A$="n" THEN 420
460 LOCATE 0,5:PRINT SP$:LOCATE 0,6:PRINT"General Write up":LOCATE 0,9,1:PRINT G$(I):A=7:GOSUB 160
470 CLS:PRINT SP$(15);P$ EDITING "Q$:PRINT:PRINT
480 FOR L=0 TO AR
490 PRINT USING"###";I: IF I$(I,L)="" THEN PRINT STRING$(30,32) ELSE PRINT I$(I,L)
490 IF PEEK(&HFA03)>19 THEN 500 ELSE NEXT:GOTO 470
500 PRINT:PRINT"Press any key to select ingredient to edit or space bar to continue selection or ESC to continue"
510 A$=INKEY$: IFA$="" THEN 510 ELSE IFA$=CHR$(27) THEN 560 ELSE IFA$="" THEN LOCATE 0,3:GOTO 480
520 LOCATE 0,20:PRINT SP$:SP$:SP$:LOCATE 0,20:INPUT"Enter no of ingredient":L: IF L<0 OR L>ARTHEN 520 ELSE LOCATE 0,21:GOSUB 330: IFA$="N" OR A$="n" THEN 520
530 LOCATE 0,20:PRINT SP$:SP$:SP$
540 LOCATE 0,20:PRINT"Ingredient "L":;:LOCATE 14,20:PRINT I$(I,L):LOCATE 14,20,1:LINE INPUT I$(I,L):LOCATE 0,21,0:GOSUB 330: IFA$="N" OR A$="n" THEN 530
550 LOCATE 0,20:PRINT SP$:SP$:SP$:LOCATE 0,20:PRINT"Any more alterations (Y/N)":GOSUB 340: IFA$="Y" OR A$="y" THEN 460
560 CLS:PRINT SP$(15);P$ EDITING "Q$:L=0
570 LOCATE 0,7:PRINT WU$(I,L):A=7:GOSUB 270: IF L>6 THEN RETURN 30 ELSE 570
580 CLS:PRINT SP$(13);P$ LISTING "Q$:PRINT:PRINT
590 FOR I=0 TO AR
600 PRINT USING"###";I: IF NM$(I)="" THEN PRINT STRING$(20,32) ELSE PRINT NM$(I)
610 IF PEEK(&HFA03)>19 THEN 620 ELSE NEXT:GOTO 590
620 PRINT:PRINT"Press any key to select recipe to view or ESC to return to menu or space bar to continue scrolling."
630 A$=INKEY$: IFA$="" THEN 630 ELSE IFA$=CHR$(27) THEN RETURN 30 ELSE IFA$="" THEN LOCATE 0,3:GOTO 600
640 LOCATE 0,20:PRINT SP$:SP$:SP$:LOCATE 0,20:INPUT"Recipe No.":I:LOCATE 0,21:GOSUB 330: IFA$="n" OR A$="N" THEN 580
650 LOCATE 0,20:PRINT SP$:SP$:SP$:LOCATE 0,20:PRINT"Printer (Y/N)":GOSUB 340: IFA$="y" OR A$="Y" THEN A=7
660 CLS:PRINT SP$(3);"Recipe";NM$(I):PRINT:PRINT: IFA=7 THEN LPRINT SP$(7);"Recipe:";NM$(I):LPRINT:LPRINT
670 PRINT"General Note:":PRINT G$(I): IFA=7 THEN LPRINT"General Note:":LPRINT G$(I):LPRINT:LPRINT
680 PRINT:PRINT"PRESS ANY KEY TO CONTINUE"
690 IF INKEY$="" THEN 690
700 CLS:PRINT"Ingredients:":FOR L=0 TO AR: IF I$(I,L)="" THEN 710 ELSE PRINT I$(I,L):NEXT
710 IFA=7 THEN LPRINT"Ingredients:":FOR L=0 TO AR: IF I$(I,L)="" THEN 720 ELSE LPRINT I$(I,L):NEXT
720 PRINT:PRINT"PRESS ANY KEY TO CONTINUE"
730 IF INKEY$="" THEN 730
740 CLS:FOR L=0 TO 7:PRINT"Instructions:":IF WU$(I,L)="" THEN 770 ELSE PRINT WU$(I,L)
750 PRINT:PRINT"PRESS ANY KEY TO CONTINUE"
760 IF INKEY$="" THEN 760 ELSE NEXT
770 IFA=7 THEN LPRINT:LPRINT:LPRINT"Instructions:":FOR L=0 TO 7: LPRINT WU$(I,L):NEXT
780 RETURN 30

```

MSX Notes -- Don Stanley

The X'Press is here -- and what a machine ! This computer is superb, I mean SUPERB. Even the documentation ! Lets briefly look at the X'Press.

The X'Press is the equivalent of a SV328 with 605A expander , costing \$1195 . However at this stage you do not get the MicroPro software, but get an extra bonus -- a built in RS232c . There is software with it -- SCHEDULE , a 5 part diary, database, time calculator, metric converter and personal info keeper. This runs under CPM. Under MSX-DOS you get 4-IN-1, a simple database/word processor/spreadsheet/disk maintenance, as well as a very good demo program, MSX-DOS utilities , MSX Disk Basic (looking practically identical to SVI Disk Basic) and various programs for setting screen color, changing editing keys and so on.

Why would anyone need a screen color changing program? SVI have done themselves ultra proud with the X'Press. It will use a TV in 80 column mode. And up to 4 colours on there as a bonus. You can use the color change program to set up foreground, background, inverse video foreground, inverse video background colours. The default is white characters on black background. YUK. The TV doesn't help this combination, it shimmers badly . I have found black characters on a grey background with black characters on a dark yellow background for inverse characters is excellent. The image is at least as good as the AMSting with its dedicated monitor. I think its better in fact -- on a TV! the only problem with this is that any time you run software under 40 columns you need to reset the colours when you go back to 80 (Both CPM and MSX-DOS boot at 40 columns by default).

What else is there that I have been able to discover in 3 days. Ah the AUTOEXEC. Under MSX-DOS you can create a file called AUTOEXEC.BAT. Then whenever you switch the system on -- it searches for this file and carries out all the commands in it. This is standard under MS-DOS and PC-DOS. Creating the AUTOEXEX file is magic. Just type COPY CON AUTOEXEC.BAT . Then anything you type from here on will go into the named file as soon as you press ^Z and ENTER (It stands for COPY from the CONsole into the disk file called AUTOEXEC.BAT). Great stuff. I have one set up so that invoking MSX-DOS leaps into 80 columns, changes the colours, sets the date (used in the disk directory) and loads my favourite Fkey definitions. Great . Any .BAT file can be invoked just by typing the file name.

More features. There are in fact 4 operating systems in here. One is CPM. Then you have MSX-DOS. From MSX-DOS you can run MSX BASIC. However it defaults to being disk basic. But hold down the shift key while starting up the system. The disk roms are locked out and hey presto its a standard MSX cassette system!. The first thing I thought was can I transfer files between them. Given that SVI have done everything else right so far surely you could. Not obvious how though. So reboot CPM and backup the disk. Format the new disk (80 tracks , 360k formatted, 3.5" disks) and type

FILECOPY. Its asking me if my source disk is CPM or MSX-DOS. In fact it asked me if my destination disk was CPM or MSX-DOS also. This utility will happily transfer from CPM to MSX-DOS and vice-versa. Since MSX Disk Basic reads ASCII files on a MSX-DOS disk -- you can transfer between CPM/MSX-DOS/MSX Basic. So you can edit long Basic programs using Wordstar send it back to Basic

Being the curious type and wanting a DDT type program under MSX-DOS I thought I'd copy DDT from CPM to MSX-DOS and see if it ran. It did up to a point, that point came when using DDT to examine a file. DDT got confused, due I suspect to different uses of page zero under CPM and MSX-DOS. As I was also unable to persuade the CPM EDITFKEY program to work under MSX-DOS properly it seems likely that most CPM software won't run properly under MSX-DOS. That's only a minor hassle, files can be transferred the other way and examined/edited/whatever under CPM. It just means a disk change and restart of the system.

Software is supplied on both the CPM and MSX-DOS disk to emulate either a VT-52 or ADM-3A terminal. With the addition of a modem this machine could function quite well as a mainframe terminal especially to DEC hardware.

Documentation. One of the features of the SV328/318 was the documentation. It was awful. Consequently I didn't open the documentation for some time. When I did the first book was called X'Press. It is the computer manual as opposed to the others which are the software manuals. Opening a page randomly revealed two boxes in grey with black print standing out among other documentation. SVI have used these standout boxes throughout to emphasise points. For instance ...

"For every disk containing the WIDTHB0 program, CHANGE.COM must be executed once (only one time) for later execution of WIDTHB0; otherwise error will occur"

Now thats the sort of thing that you could guarantee the 318/328 manuals would have left out. The most important thing about a program, emphasized and clearly written. And thats the whole manual in a nutshell. It really is superb, even down to the stage of telling you how to run games that are too big for MSX Disk Basic. Furthermore, everything is set out in numbered stages. I have followed exactly the instructions for disk formatting, changing function keys and sysgening. NO ERRORS at all exist in that part of the documentation. I haven't found any errors yet in fact anywhere.

What of the other manuals. The CPM manual looked a bit thin. Maybe they slipped up and used ultra small print. I guess my opinion of the 318/328 documentation is saying something must be wrong somewhere. Opening the manual reveals .. a CPM manual written for humans, not the machines at Digital Research. This is just the sort of manual that should have been written for CPM years ago. It explains what CPM is about, what files are, clearly explains all the CPM commands, its got lots of examples and the

only really technical information is the list of BDOS function calls on the very last page. This is the first edition , JULY 1985 print. The standard Digital Research manual is available as an extra.

The MSX-DOS manual just carries on the same way. Very clear, concise, highlights in black on grey, technical details of how the MSX-DOS disks look (complete specs on file allocation tables, memory map , disk formats file control block , I/O Address map). Very clear descriptions of the commands, backing up and so on.

The Disk Basic manual even remembers to put in COPY and DSKF. Again it is clear and concise, with little evidence of Asian to English translations. Then there is the MSX BASIC manual, not much more than the standard copy of the MSX manual from Microsoft. but again with highlighting of important points and also some of the command descriptions have been rewritten to clarify them. Theres just no way that you could ever believe this company put out the 318/328 manuals. These manuals are top notch (And of course are complimented by the many books now available with technical MSX documentation).

I'm desperately trying to find something wrong with the manuals. They can't be this good. Computer manufacturers are notorious for remembering they have to write manuals the day before they ship the systems. Believe me, as well as setting a new 8 bit standard of excellence in the hardware, SVI have done their utmost to do the same for the documentation. If I'm raving about this, its because I have never seen such superb documentation on a home/small business computer.

What does the X'Press look like. Its light. I know that because I carried it home a mile from the post office. When you open the box you see an elegant black carrybag and 2 pieces of polystyrene. Thoughtfully , the bags handle faces you. grip the handle and pull out the bag, unzip and there it is. No mess of cables in there, just a computer and 2 disks, with a power cable and transformer as well as TV cable nicely sitting at the back.

Removing the computer reveals no obvious plugs except a cassette port and 2 joystick ports. On the top are two LED's , one for when the system is switched on and one for when the disk drives are in use. A further LED sits on the CAPS LOCK key. Turning it over reveals a pull out handle which protects all the slots at the rear as well as providing some elevation when the machine is sitting on a desk.

What are all these holes at the back for. Theres a power cable connector. Good start. Theres a printer port. Theres a 2nd disk drive port. Theres an RS232c port. Theres a TV slot. Theres a monitor port. Theres a speaker port. And an ON/OFF switch. All this and superb documentation for \$1195. And of course the MSX cartridge slot on the top. And don't forget the built in 3.5" disk drive.

Thats your 738. SVI have cut no obvious corners, and certainly have come up with an elegant and superbly documented computer. With the MSX standard supplying software by the hordes and CPM right there when you buy the system, the X'Press is already very well endowed with software. There are masses of peripherals, including joysticks, disk drives, cassette, "stringy floppies" which seem suspiciously like microdrives and so on. For \$1195 this is the computer bargain of the decade.

READING MACHINE CODE TAPE HEADERS.

BY David Kelly

With a machine code program which has been BSAVE'd, making a back-up copy is difficult unless you know where abouts it resides in memory. The program detailed below will read and print the start, end and execution addresses for any BSAVE'd program, making the copying task much simpler.

```

5 REM BIN RD.      By D.Kelly
10 AD=&HD000
20 READ D$
30 IF D$="*" THEN 190
40 POKE AD,VAL("&H"+D$)
50 AD=AD+1
60 GOTO 20
70 DATA 0E,D0      :REM LD C,D0
80 DATA CD,34,1F    :REM CALL SRCCAS
90 DATA CD,97,1F    :REM CALL 1F97
100 DATA CD,3A,20    :REM CALL CSRDON
110 DATA CD,76,1E    :REM CALL 1E76
120 DATA 22,30,D0    :REM LD (TEMP1),HL
130 DATA CD,76,1E    :REM CALL 1E76
140 DATA 22,32,D0    :REM LD (TEMP2),HL
150 DATA CD,76,1E    :REM CALL 1E76
160 DATA 22,34,D0    :REM LD (TEMP3),HL
170 DATA C9         :REM RET
180 DATA *         :REM End of data
190 FOR F=&HF99E TO &HF9A3
200   POKE F,32
210 NEXT F
220 DEFUSR=&HD000
230 A$=USR(0)
240 PRINT
250 AD=&HD030
260 PRINT "START: ";
270 GOSUB 500
280 PRINT "END : ";
290 GOSUB 500
300 PRINT "EXEC : ";
310 GOSUB 500
320 PRINT
330 END
500 PRINT HEX$(PEEK(AD)+256*PEEK(AD+1))
510 AD=AD+2
520 RETURN

```

PROGRAM OPERATION :

The program operation is fairly simple since some of the ROM routines are utilized to do all the work. A breakdown of the machine code section is as follows :

Line 70 loads the C register with D0, which is the identifier for a BSAVE'd program.

Line 80 calls the ROM routine which performs several functions:-

- a) Check PLAY is depressed on the recorder.
- b) Wait until it finds a 'leader tone', and the synch character (see newsletter #13).
- c) Read the header to see if it has found the correct file (file type and file name).
- d) If it has found the correct file then return, else go back to step a).

Upon return from this routine, HL is pointing to the 'Found:' message and the name of the file is in locations F9A7 to F9AC

Line 90 calls a routine which prints out the 'Found:' message, and the name of the file.

Line 100 calls a routine which performs the same operation as steps a) and b) in Line 80 (ie. it waits for the 'leader' of the file body).

Line 110 reads 2 bytes from the cassette, and stores them in HL. These first 2 bytes are the start information.

Line 120 stores this in a temporary location.

Lines 130-160 read 4 more bytes from the cassette and stores them in temporary locations. These correspond to the end and execution information.

Line 170 returns back to BASIC.

The BASIC section of the program is straightforward, however there is several points to note :

- a) In lines 190 to 210 the file name is set to 6 spaces. This prompts the routine at 1F34 to read the first machine code file it comes across.
- b) In the routine at 1F97, the 'Found:' message is treated as a string. This is returned to BASIC as the result of theUSR. This is the reason for calling the machine code program with A\$=USR(0).
- c) Note that the addresses printed are in Hexadecimal. If you prefer them in decimal, change line 500 to :
500 PRINT PEEK(AD)+256*PEEK(AD+1)

PROGRAM USE :

- 1) Check the program is entered correctly (especially the machine code DATA statements).
- 2) CSAVE the program.
- 3) RUN the program.
- 4) PLAY the tape with the BSAVE'd program on it, at the correct point.
- 5) Make a note of the addresses printed by the program. If there is more than one BSAVE'd program on the tape it may be more convenient to get the addresses for these programs at the same time. Change line 330 to GOTO 230
- 6) BLOAD the program.
- 7) When it has finished loading, BSAVE it onto the backup tape (or Disc) using the addresses noted in step 5).

PUBLIC DOMAIN SOFTWARE

S. KN:

ays

There are now 8 Public domain tapes labelled 1 - 9 with No.2 deleted. Full contents descriptions of tapes 1,3,4,5 are in the June 85 issue of this Newsletter. For those who have recently subscribed a brief description of contents is given below.

TAPE 1 Twentyone Games programs from the Australian Users Group
Includes; Mathematics, Card games, Music, Adventure Games
Hangman, World Map, Disassembler, Graphics, Skill programs

TAPE 3 Eighteen Programs:
Mainly Disk Utility programs. Star Trek Game, CPM Utility
State -running a country, Eisection, Riteman dump, 3Dplot

TAPE 4 Nineteen Programs:
FRED -Disk utility, Machine code loader / disass, Graphics
Yatzie, Disassembler, Adventure Game, Graphics, Utilities

TAPE 5 Eighteen Programs:
Educational Program entries; Futuristic Voyage, Sprites
Blackjack, Graphics Utilities, Music, Racing car, Hex dump

TAPE 6 Sixteen Printer based programs:
Two High speed Normal/Double density Screen Dump programs
Scrolling screen Text Editor with up to 255 columns
Disassembler II by Don Stanley for Memory / Disc / Tape
Biorhythm analysis, Human Compatibility, Calendar - 1999
Cassette inlays for Music and Computer program labelling
Statistics -graphs and analysis, Bar Graph 40 variables
GPHipt Prints your fonts to printer, Z80plu Peek - Poking
Format re-formats ascii saved program, blist -M/locations
WORDS - Words puzzle - works only to printer.

TAPE 7 Twelve Games programs from Australian Users Group:
Four adventure games ... two of these are only for SV328
Two Puzzles and the rest skill type games.

TAPE 8 Sixteen Utility programs from the Australian Users Group:
Three Graphics artist utility programs, 8 * 8 Font Editor
SOUND Command Utility, Disassembler, Assembler & Display
Text Editor, Algebra Maths, Cube rotation, Font chr set

TAPE 9 Fourteen assorted programs:
Alcohol Absorption & Metabolism, Othello, World Cup soccer
Patience card game, Three graphics demos, Reaction timer
Life Expectancy, Screen message display, Mon. peek/poke
Graphics mode one display, Inverted character set, Format

Software can be ordered from; PD Software, P.O. Box 26050
Newlands, Wellington, New Zealand

Costs: Each tape is \$3 each with a handling charge
of \$2.50 for each order made. Make Cheques
payable to G. Clark.

Note: Programs from this list are broadcast on a Access Radio ..
on the 1st Saturday of every month at 9.00am on station
2YB 783 kHz.

```

10 REM FATHER CHRISTMAS / TREE
20 REM DECK THE HALLS * G.CLARK NOV 85
30 COLOR 15,0,0:SCREEN1,3
40 ONSTOPGOSUB280:STOPON
50 CO=2:GOSUB290:GOSUB210
60 FORX=1TO3:LOCATE30+X,15:PRINT" M E R R Y"
70 LOCATE3+X,35:PRINT"C H R I S T M A S":NEXT
80 FORX=1 TO15:GOSUB190:Y=Y+2:NEXT
90 FORX=10TO30:GOSUB190:Y=Y+2:NEXT
100 FORX=20TO45:GOSUB190:Y=Y+2:NEXT
110 FORX=40TO60:GOSUB190:Y=Y+2:NEXT:CO=7:X=3
120 FORY=YTO172:GOSUB190:Y=Y+1:NEXT:CO=9
130 FORX=15TO10STEP-.5:GOSUB190:Y=Y+2:NEXT
140 REM MESSAGES PUT HERE
150 PLAYA$,B$,C$:GOSUB370:PLAYA$,B$,C$:GOSUB370
160 PLAYD$, ,F$:GOSUB370:PLAYA$,B$,C$:GOSUB370
170 IFINKEY$="" THEN170
180 GOTO150
190 LINE(120-X,Y )-(120+X,Y ),CO,BF
200 LINE(120-X,Y+1)-(120+X,Y+1),CO+1,BF:RETURN
210 REM MUSIC
220 A$="T255L305C04L8B-L4AGFGAFL8GAB-GL3AL8GL4FEL2F"
230 B$="T255L304AL8GL4FEdeFCL8EFGL3FL8DL4CCL2C"
240 C$="T255L303FL8EL4FCDCFFL8CCCL3FL802B-L403CCL2F"
250 D$="T255L304G04L8AL4B-GL3AL8B-05L4C04GL8AB-05L4CL8DEL4FEDL2C"
260 F$="T255L303CL8CL4CCL3FL8FL4FEL8CDL4CL803B-04CL4CC03B-L26"
270 RETURN
280 COLOR 15,4,5:SCREEN0,1:STOP
290 FORA=1TO32:READVV$:V$=V$+CHR$(VAL("&H"+VV$)):NEXTA
300 SPRITE$(28)=V$
310 FORA=1TO32:READWW$:W$=W$+CHR$(VAL("&H"+WW$)):NEXTA
320 SPRITE$(29)=W$
330 FORA=1TO32:READXX$:X$=X$+CHR$(VAL("&H"+XX$)):NEXTA
340 SPRITE$(30)=X$
350 FORA=1TO32:READY$:Y$=Y$+CHR$(VAL("&H"+YY$)):NEXTA
360 SPRITE$(31)=Y$:RETURN
370 IFA=0THEN400
380 A=0:FORX=1TO80:GOSUB420:NEXT
390 A=0:RETURN
400 FORX=80TO1STEP-1:GOSUB420:NEXT
410 A=1:RETURN
420 PUTSPRITE28,(220-X,127), 6,28
430 PUTSPRITE29,(220-X,127),15,29
440 PUTSPRITE30,(220-X,159), 6,30
450 PUTSPRITE31,(220-X,159),15,31
460 RETURN
470 REM HEAD RED
480 DATA 00,C0,E1,03,E0,E3,E0,E1,E0,E0,FC,FC,7E,3E,0F,0F
490 DATA 3B,F8,F8,F8,00,F8,00,20,80,80,0C,0E,1F,1F,37,F7
500 REM HEAD WHITE
510 DATA 00,00,00,E0,03,00,03,02,03,03,03,03,01,01,00,00
520 DATA 06,06,00,00,F8,00,F8,D8,78,78,F0,F0,E0,E0,C0,00
530 REM LEGS RED
540 DATA 1F,1F,3F,3F,00,00,1E,1E,1E,1E,1E,1E,00,00,00,00
550 DATA FF,FF,FF,F8,07,03,F0,F0,F0,F0,F0,F0,00,00,00,00
560 REM LEGS WHITE
570 DATA 00,00,00,00,7F,7F,00,00,00,00,00,00,1C,1C,7D,7D
580 DATA 00,00,00,07,F8,F8,00,00,00,00,00,00,70,70,F0,F0

```

Life, the Universe and CP/M

Part: the one that comes after two.

It's amazing. I have had three phone calls in the last week asking how to make the CP/M system autobooting (start to run a program automatically on startup).

There is quite a simple solution really, and Don has written a BASIC program to do it, with some machine code imbedded in it. So this month Don is providing the work, and me the words.

There is a public domain CP/M program called autoboot or autostart or something like that, that is available from the New Plymouth users group, but it needs adapting to the SV system, and as far as I know, no one has done it yet.

First we have the background, the bit where I get to spout off about what happens when you turn on your SV with a CP/M disk in the drive.

The ROM in the SV goes through some checks, working out how much RAM is available and things like that, (if you want to know more, see Don), and then checks if there is a disk controller plugged in. If not the system goes on to ROM BASIC and the rest of this article is wasted.

If there is a controller, the ROM checks if there is a disk, and if so, the commands are given to load the first sector of the first track of side 0 into memory at C000 hex. The ROM then gives control over to what ever program is at that location. If the disk was a SV Disk BASIC one, the program there loads the BASIC into high memory, does some more checks, and then returns control to the ROM to finish the startup and display the logo etc. But as it was a CP/M disk this is what happens.

The code starting at C000 hex loads the CCP, BDOS and BIOS into high memory, and then jumps to location E600 hex, which is another jump to a location higher up which is called CBOOT. CP/M has two types of startup: WARM and COLD. The COLD start is executed when the system first starts up, i.e. when you first turn it on, or after you press the reset button, which is effectively that same thing. The WARM start should be executed at the end of every program that runs. What it does is reset the system tables ready to run a new program. I said should because some programs do not exit properly, through the prescribed channels, and this can tie the system up at some later time.

The COLD start routine does those things that should only have to be done once for each time you run CP/M, like switching out ROM BASIC, setting up the RS232 card, the 80 column card and things like that. Then the COLD start routine falls through to a location in the middle of the WARM start routine, called WBOOT1.

This is the part of the WARM start routine that has to be executed for a cold and a warm start, and both routines share the code. The first part of the WARM start routine, which starts at (believe it or not) WBOOT, does only those things that have to be done for a WARM start.

After both the COLD and WARM start routines have done their things, control is passed to the CCP. One of the things the CCP does is check its command input buffer. This is where we get interested. The command input buffer is located at location CCP+7 (which means the location given by the start of the CCP plus seven bytes), and consists of:

a byte counter to tell how many characters follow and 16 bytes for characters to go in.

This buffer is followed immediately by the CP/M copyright message, but not the one that you see on the screen when CP/M starts.

CCP looks at this buffer and if the first location (the counter at CCP+7) is not zero it interprets the characters in the buffer up to the first null (value 00) as a file name, and tries to find and run it. It is just as if you had typed something yourself. It is this feature we are going to exploit.

If you want CP/M to execute a program every start, both WARM and COLD, all you have to do is change the value at location CCP+7 to how long the command is, and put the command into the buffer, null terminated (the last character is a null). The program will then run when you turn the machine on, every time you use control-C, and at the end of most other programs. This can get a bit annoying.

What we have to do is patch CP/M so that the program is only run on a COLD start. We add a bit of extra code that checks if it is a WARM or COLD start, and if it is a WARM start, it changes the counter byte to zero so the CCP will not execute the program.

Here is the order:

- Patch the CCP so that the program name is there, and will run every start (WARM or COLD).
- Write a small addition to the COLD start routine that sets a flag in high memory to indicate that it was a COLD start.
- Write a small addition to the WARM start routine, or the CCP program (in Don's case, he has use the memory taken by the copyright message in the CCP) that checks for that flag, and if it is set, it resets it. This is to ensure that next time, the WARM start does not see it again. If the flag is not set (it must be a WARM start), the byte counter at CCP+7 is set to zero so the program will not run. This small routine then returns to the normal code.

What all this means is that on a WARM start, the program will not run, but on a COLD start it will. That sounds suspiciously what we wanted, does it not?

On the SV, the location of the CCP is at D000 hex, but it is no good patching in memory because you cannot save it out to disk. You either have to patch on the disk, which is what Don's program does, or use SYSGEN to get the operating system into lower memory and patch there, then continue with SYSGEN to write it out again.

Anyway that is enough for this time of the year. Have a merry christmas all, and all the very best for you and yours for the next year.



Don's program to patch CP/M as explained by Ross above is on the following pages. It is written in Basic for easier access to the disc sectors. Simply type it into basic, save it to a basic disc and then test it on a spare CP/M system disc. When running the program, any part of it can be avoided by pressing any key but "ENTER" at the first prompt.

Parts of the program have been included in various forms in previous newsletters: CP/M.zap in #11, disk spin off patch in #9, and the Autoboot function is described above. They are here presented in a BASIC program which is easy to type in (if you leave out the REM statements) and easy to follow.

This program is Public Domain. That means it can be used by anyone, given to anyone, but not sold to anyone or used in any program which will be sold.

CP/M Alter Program - note runs under 80 columns

```
1000 WIDTH80:SCREEN0,0
1010 REM CPM AUTOBOOT and FUNCTION KEY and DISK SPINOFF change program
1020 REM
1030 REM W A R N I N G --- W A R N I N G --- W A R N I N G
1040 REM
1050 REM This program has been successfully tested on CPM2.20,2.22 & 2.23 with
1060 REM the word REVISION spelt REVERSION. It works fine on these versions, but
1070 REM I suggest you run the thing on a spare sysgened CPM disk first to make
1080 REM sure it works. For the 2.23 version with REVISION spelt correctly try
1090 REM changing line 340 so the 55 is 54 or 53 or 52 or 56 or 57. I think 54
1100 REM should be the right one. Everything else for that version SHOULD be OK
1110 REM but try it on a spare disk anyway at first until you see what it does
1120 REM
1130 REM Don Stanley -- 1985--November--Sunday--24th--Raining--Yuk
1140 REM
1150 REM CPM 2.20/22/23--Track 1 Sector 13 Bytes 8-23 can be used for Autoboot
1160 REM First Byte is numbered 1 not 0. Byte 8 should be non zero to
1170 REM Autoboot. Autoboot must end with a null or will drift into
1180 REM the patched BIOS
1190 REM
1200 REM This program AUTO Detects which version of CPM is being
1210 REM used. It uses the bytes at Track 1 Sector 2 which are the
1220 REM version number on standard CPM (Bytes 27,28,29,30) For 2.22
1230 REM or at Track 1 Sector 1 (Bytes 58,59,60,61) for 2.20
1240 REM
1250 REM PUBLIC DOMAIN -- under NO CIRCUMSTANCES WHATSOEVER may this program
1260 REM be incorporated into non-public domain software
1270 CLS
1280 LOCATE 5,5:PRINT" Insert CPM Disk in drive 1"
1290 LOCATE 5,7:PRINT"No WRITE PROTECT TABS Please !"
1300 LOCATE 5,9:PRINT" Press Any Key When Ready ":IF INKEY$="" THEN 1280
1310 :
1320 REM Get Version Number
1330 CLEAR 1000
1340 A$=DSKI$(1,1,2) ' is it 2.22 or 2.23
1350 VN$=MID$(A$,27,4)
1360 IF VN$="2.22" THEN 1430
1370 VN$=MID$(A$,55,4) ' for the REVERSION version !
1380 IF VN$="2.23" THEN 1430
1390 A$=DSKI$(1,1,1) ' is it 2.20
1400 VN$=MID$(A$,58,4)
1410 IF VN$="2.20" THEN 1430
1420 PRINT "CPM Disk Is Unknown version -- Aborting":END
1430 :
1440 REM We have the version Number -- Now Set Up Alter screen
1450 CLS
1460 PRINTSTRING$(39,"="):PRINT:
1470 PRINT:PRINT"CPM Version ";VN$;" System Disk Update"
1480 PRINT:PRINT"AutoBoot Setup":PRINT:PRINT"Press Enter to Continue"
1490 A$=INPUT$(1):IF ASC(A$)<>13 THEN 1770
1500 DATA 3a,7a,fe,fe,0,3e,ff,32,7a,fe,ca,5c,d3,3c,32,07,d0,c3,5c,d3,*,*
1510 :
1520 REM Data patches BIOS to prevent autoboot on warm start--2.20/2.22/2.23
1530 :
1540 REM program is patched into CCP at locations d03f--d052
1550 REM bytes at d000 are changed from 5cd3 to 3fd0
1560 REM LD A,(FE7A)
1570 REM CP 0 ! LD A,FF
1580 REM LD (FE7A),A ! JZ D35C
```

```

1590 REM INC A : LD (D007),A : JC D350
1600 :
1610 IF A$ ( ) "*" THEN READ A$:MC$=MC$+CHR$(VAL("&n"+A$)):GOTO1610
1620 PRINT"Enter UP TO 15 Chars For AUTOBOOT":PRINT:PRINT"UpperCase ONLY":INPUT
AB$
1630 IF LEN(AB$) < 15 THEN AB$=AB$+STRING$(16-LEN(AB$),0) ELSE AB$=MID$(AB$,1,15)+
CHR$(0)
1640 PRINT"Press Enter To Confirm":Z$=INPUT$(1)
1650 IF ASC(Z$) < 13 THEN 1480
1660 :
1670 REM Autoboot Done - now save it -- 2.22/2.20/2.23 use same disk spot
1680 A$=DSKIS(1,1,13):AB$=CHR$(4)+AB$:MID$(A$,64,LEN(MC$))=MC$
1690 MID$(A$,8,17)=AB$:MID$(A$,2,2)=CHR$(&H3F)+CHR$(&HDO)
1700 FIELD #0,128 AS X$,127 AS Y$
1710 LSET X$=LEFT$(A$,128) : LSET Y$=RIGHT$(A$,127)
1720 DSKO$1,1,13
1730 :
1740 REM Autoboot is now saved !
1750 REM Now hit the Function Keys
1760 :
1770 CLS:PRINT"Function Key Setup":PRINT:PRINT"Press Enter To Continue"
1780 A$=INPUT$(1):IF ASC(A$) < 13 THEN 2570
1790 :
1800 REM First Get a function Key String
1810 PRINT "Please Wait a tick "
1820 IF VN$( ) "2.22" AND VN$( ) "2.23" THEN 1900
1830 A$=DSKIS(1,0,2)
1840 FK$=MID$(A$,118,11)
1850 A$=DSKIS(1,0,3)
1860 FK$=FK$+MID$(A$,1,128)
1870 A$=DSKIS(1,0,4)
1880 FK$=FK$+MID$(A$,1,21)
1890 GOTO 2000
1900 IF VN$( ) "2.20" THEN 1970
1910 A$=DSKIS(1,0,2)
1920 FK$=MID$(A$,103,25)
1930 A$=DSKIS(1,0,3)
1940 FK$=FK$+MID$(A$,1,128)
1950 A$=DSKIS(1,0,4)
1960 FK$=FK$+MID$(A$,1,6)
1970 A$=DSKIS(1,0,2)
1980 :
1990 REM heres where the work is done getting the new key defs
2000 CLS:PRINT"Key # Current Def"
2010 FOR I= 1 TO 10
2020 LOCATE0,I:PRINTUSING" ## \ \":I,MID$(FK$,16*I-15,16)
2030 IF INSTR(FK$,CHR$(13)) < 1 THEN LOCATE 35,I:PRINT" (cr) ";
2040 NEXT
2050 PRINT:PRINTSTRING$(39,"=")
2060 LOCATE 2,15:PRINT"Enter # to Change (0=Quit)";
2070 KN=VAL(INPUT$(2)):IF KN=0 THEN 2230
2080 IF KN>10 OR KN<1 THEN 2060
2090 LOCATE2,17:PRINTSTRING$(200," ")
2100 LOCATE2,17:PRINT"Enter New Key Definition-KEY ";KN:KY$=""
2110 LOCATE2,19:INPUT KY$:IF LEN(KY$)>16 THEN KY$=LEFT$(KY$,16)
2120 LOCATE2,17:PRINT"New Function Key Will Be ":LOCATE2,19:PRINTKY$"
2130 LOCATE2,21:PRINT"Press Enter To Confirm Change":A$=INPUT$(1)
2140 LOCATE2,17:PRINTSTRING$(200," ") :IF A$ < 13 THEN 2060
2150 A$="n":IF LEN(KY$)=16 THEN 2180
2160 LOCATE2,17:PRINT"Press Enter For A Carriage"

```

```

2170 LOCATE2,18:PRINT"Return At End Of Key":A$=INPUT$(1)
2180 LOCATE9,KN:PRINTSTRING$(29," " ) ;
2190 LOCATE9,KN:PRINTKY$:IF A$(<)CHR$(13) THEN2200ELSEKY$=KY$+CHR$(13):LOCATE35,KN:
PRINT"(cr)";
2200 IF LEN(KY$)(<)16 THEN KY$=KY$+STRING$(16-LEN(KY$),0)
2210 MID$(KY$,16,1)=CHR$(0) ' ensure last bit is zero to prevent massive keys
2220 MID$(FK$,16*KN-15,16)=KY$:GOTO2060
2230 IF KY$="" THEN 2570
2240 :
2250 REM Save them on the system tracks
2260 IF VN$="2.20" THEN2420
2270 A$=DSKIS(1,0,2)
2280 MID$(A$,118,11)=MID$(FK$,1,11)
2290 FIELD #0,128 AS X$,127 AS Y$
2300 LSET X$=LEFT$(A$,128):Y$=RIGHT$(A$,12)
2310 DSKO$1,0,2
2320 A$=DSKIS(1,0,3)
2330 MID$(A$,1,128)=MID$(FK$,12,128)
2340 LSET X$=LEFT$(A$,128):LSET Y$=RIGHT$(A$,127)
2350 DSKO$1,0,3
2360 A$=DSKIS(1,0,4)
2370 MID$(A$,1,21)=MID$(FK$,140,21)
2380 LSET X$=LEFT$(A$,128):LSET Y$=RIGHT$(A$,127)
2390 DSKO$1,0,4 : GOTO 2570
2400 :
2410 REM 2.20 specific guff here
2420 A$=DSKIS(1,0,2)
2430 MID$(A$,103,26)=MID$(FK$,1,26)
2440 FIELD #0,128 AS X$,127 AS Y$
2450 LSET X$=LEFT$(A$,128):Y$=RIGHT$(A$,12)
2460 DSKO$1,0,2
2470 A$=DSKIS(1,0,3)
2480 MID$(A$,1,128)=MID$(FK$,27,128)
2490 LSET X$=LEFT$(A$,128):LSET Y$=RIGHT$(A$,127)
2500 DSKO$1,0,3
2510 A$=DSKIS(1,0,4)
2520 MID$(A$,1,6)=MID$(FK$,155,6)
2530 LSET X$=LEFT$(A$,128):LSET Y$=RIGHT$(A$,127)
2540 DSKO$1,0,4
2550 :
2560 REM switch off the disk spin
2570 CLS:PRINT"CPM Disk Spin Stop "
2580 PRINT:PRINT"Press Enter to Alter BIOS"
2590 A$=INPUT$(1):IF A$(<)CHR$(13) THEN 2710
2600 IF VN$="2.22" OR VN$="2.23" THEN A$=DSKIS(1,1,1)
2610 IF VN$="2.20" THEN A$=DSKIS(1,0,17)
2620 IF VN$="2.22" THEN MID$(A$,58,2)=CHR$(&H32)+CHR$(0)
2630 IF VN$="2.20" THEN MID$(A$,94,2)=CHR$(&H32)+CHR$(0)
2640 IF VN$="2.23" THEN MID$(A$,73,2)=CHR$(&H32)+CHR$(0)
2650 FIELD #0,128 AS X$,127 AS Y$
2660 LSET X$=LEFT$(A$,128) : LSET Y$=RIGHT$(A$,127)
2670 IF VN$="2.23" OR VN$="2.22" THEN DSKO$1,1,1
2680 IF VN$="2.20" THEN DSKO$1,0,17
2690 :
2700 REM check it ?
2710 CLOSE:CLS:PRINT"Boot CPM For Check ?? (Y/N)"
2720 A$=INPUT$(1):IF A$="Y" OR A$="y" THEN DEFUSR=0:A=USR(0)
2730 WIDTH80:END

```

ACTION COMPUTERS
P.O Box 7442
Wellington South
Wellington
New Zealand

18/11/85

Dear SVI / MSX Owner , User Group

ACTION COMPUTERS is a recently formed software house offering support to SpectraVideo and MSX computers. We are interested in seeing any software which you have developed which you may wish to consider marketing.

Our royalties are very competitive and based on a sliding scale which means that as more copies of your program are sold the royalty increases.

Generally software distribution will be by tape and the retail price will be dependent on the software. Generally this will be \$24.95. All sales will be by mail order at this stage.

We are interested not only in Basic/MC software but also CPM software. Thus if you have developed any application in CPM which is not currently available we would be interested in this. We will consider any form of software, games, educational, utilities etc. We are particularly interested in TURBO PASCAL utilities and software written in TURBO. Any software written for MSX-DOS we would also be interested in evaluating.

ACTION COMPUTERS will convert your SVI software to MSX and vice versa wherever possible. Thus the potential sales become much larger.

We are interested in marketing software for other machines also. If you are aware of any software developers who are looking for an outlet for their software, please pass our address on to them.

If you have any queries about ACTION COMPUTERS please do not hesitate to write to us. Enclosed is an order form with details of 5 SVI titles now available. We would appreciate it if users groups could print this information in newsletters and also inform members at meetings. All titles except SUPER ART are available for both SVI and MSX, SUPER ART will be available for MSX Disk Basic systems shortly. Please allow at least 2 weeks for receipt of orders, MSX copies may take a little longer at this

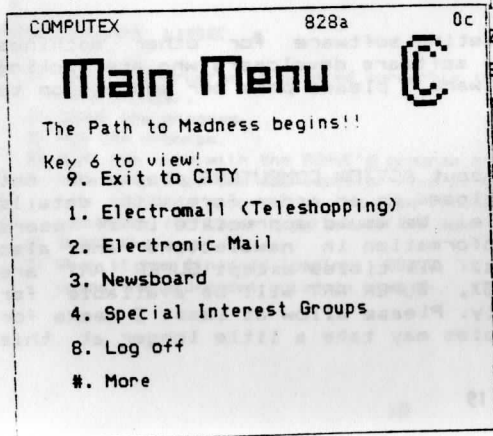
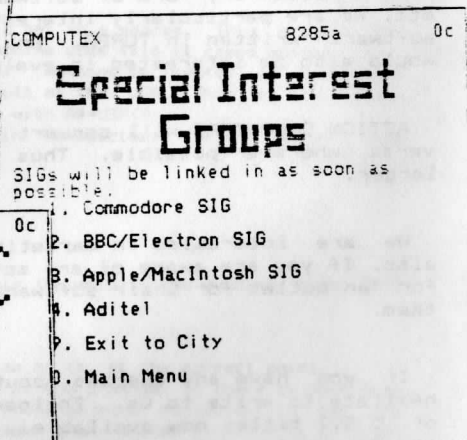
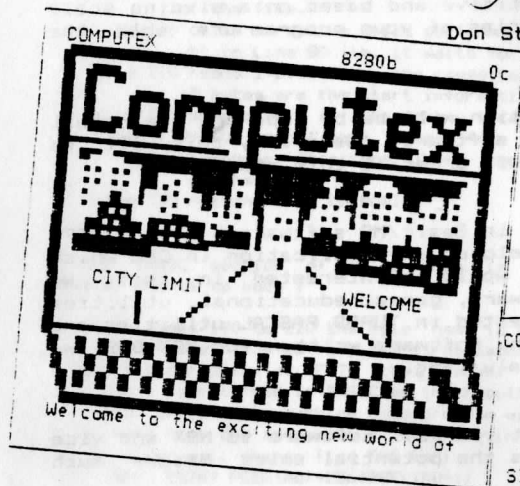
stage. For groups with newsletters we would be interested in advertising in these if you could send details of advertising rates.

As well as software development and sales for the micro field, ACTION COMPUTERS are specialists in SAS (both mainframe and PC versions), as well as other mainframe software. We carry out contract programming in SAS, as well as contract work in the micro field, where we offer expertise in MACHINE CODE, PASCAL, DBASE2, CPM, as well as other less known software. We can also offer training for new buyers of SVI/MSX hardware in the Wellington area.

Yours Faithfully

Don Stanley

Don Stanley, Programmer/Consultant



PROMISING COMPUTEX SVI/MSX.

from Colin Chin

Following the interest created by my article in last month's Newsletter, Brian Wilson (Auckland) was a welcome visitor to the November meeting. His address brought the SVUG (Wgtn) up-to-date, with the BBS scene in Auckland - and the pitfalls created on the N.Z. computer scene, particularly from hackers. (Thanks Brian, this info. was most welcome - especially with recent press comments from P.O. officials).

However, it was on the Videotex scene that SVI users were bought up-to-date. Meantime, I've also been fortunate enough to be able to review COMPUTEX, but to clarify a few points for those who may be puzzled.

1. SV318/328 do not have a suitable modem available in N.Z. This is despite requests to CDL for the modem that "came out in the beginning of 1984"...quoted from John Palmgren's GRAPHIC SOUND & SPRITES ON THE SPECTRAVIDEO (page 120) distributed in NZ by CDL.

2. The suitable modem to connect Videotex is not only to be 1200/75 split baud rate, but also to be in Prestel-format.

3. Any modems connected to the Post Office lines must also be NZ Post Office approved, otherwise under P.O. regulations, they can be confiscated.

4. There are now over 2000 users of Videotex in NZ, mainly in the business sector. Users are mainly farmers, financiers, travel agents, small homebank customers, small investors. The information available is dependent upon the databases, and naturally upon demand. These are now being enhanced by PC users with the addition of COMPUTEX. This list is expected to grow to at least 6000 by the end of 1986-87.

5. A suitable modem and software is available for MSX (marketed in the UK), which can be used for SV318/328 if the I/O ports are relocated accordingly. Brian showed the meeting a cable suitable for interfacing the MSX with SV but the M/C software he had devised was not bug-free. Any offers, programmers?

6. Brian advises he has a copy of the MSX software, and is hoping to be able to convert it for SV users, shortly - to market it together with his interface cable. The latter being only in the prototype stage.

7. Besides the modem, to connect your terminal or computer to the phone line, you will need a BT jackpoint, which all new PERT telephones do have. Installation costs are a one-off \$45.

8. The N/L Editor now advises me that the Brisbane SVUG have started a BBS service, whilst an Otago SV/MSX fan has marketed a CP/M disk for operating BES. Although there are several PDS programs ie X-Modem, Yam7, etc. I do not have sufficient knowledge of CPM to comment on their usefulness to SV, even though one sold by the ASUG (Tasmania) has been formatted for SV.

COMPUTEX

Here are some sample pages from Computex. Since the last meeting, Garry Clark has offered to design some suitable logo

pages out of the four free frames accepted by the SVUG (Wgtn). Hopefully, we can get in at No5 or 6, on the SIG menu.

In response to my article, Laurie Callender has written on behalf Taranaki U.G. asking which modem is suitable - the preceding comments answer this. His other comment was that Taranaki would be happy to support a Joint Registration on Computex. C'mon you other groups - we may be able to either regularly teleconference if we all chip in.

I've gone through several of COMPUTEX pages, and it looks very promising. I was disappointed that DITS & BYTES articles were still to be brought onto screen - admittedly, I was looking for the latest on the X-press. However, once the latest PC news is unloaded direct from overseas, it'll be the biggest PLUS as far as new developments go.

Spectravideo is not entirely left out, if you know what to look for. For instance, MICROSTYLE COMPUTERS LTD were the only SV dealer advertising. However, it's great for out-of-towners wanting to purchase SV items without a toll call. Teleshopping at its best. Hopefully, CDL will advertise all the SVI/MSX goodies, instead of us users having to scabble around our old tattered price lists to see if that Coleco adapter is available, at what price?

Another discovery was amongst the Computex subscribers. I spotted Bob Curwood's name - a regular contributor to our Newsletter, and a member of our Christchurch SVUG. I've sent him an electronic message, as I tried to ring him last week without success, so I know the message will be awaiting him when he next dials into Computex. Because there's no reply, I presume he's out of town. But, with Videotex, you can dial from anywhere in N.Z. if one chooses, providing you can find another terminal.

One final comment, Videotex has over several million users in France, because the Post Office there supplied it free to all telephone subscribers.

Declaration of Interest : I am a shareholder in a company Douglas Chin Ltd, who have recently been appointed Agents for Aditel. However, this appointment does not colour personal enthusiasm for Videotex which was more than useful during both the recent Journalists and Bank Officers strikes. My normal banking transactions, and access to news information was available throughout both these strikes. Naturally, if any information on Aditel/Computex is required, I'd be only too pleased to assist.

GLOSSARY

The following has been subverted from correspondence to the Group which your Editors found interesting.

ABORT - When the bus is full and the conductor shouts "All Abort!"

ABSOLUTE ADDRESS - Where she really lives, not where she told you she lives

ACCESS - A large painful boil on the chips from too much hacking

A/C CONVERTER - A physician specialising in bi-directional surgery

ACCURACY - Something impossible for programmers to obtain

ADDER - The part of the cow that counts how much milk is left

ADDRESS - What binary hackers wear when they aren't wearing a suit

ALGOL - What you sneak into the punch at a computer party to live it up

ALGORITHM - An unsteady gait as a result of too much ALGOL

ANALOG - Banana and icecream treat

AMBIGUITY ERROR - When something is definitely wrong, ..maybe?

ARRAY - Spontaneous exclamation by a computer user when something works for a change; rarely used

ASCII - The key to your girlfriend's apartment

AUTOMATIC CHECK - The one that you write to your software supplier every month

AUTO ANSWER - Similar to Dial-A-Prayer, but instead of a clergyman, you get Ted Computer

AUTO REPEAT - A key, which when held down, when held down

AUXILIARY MEMORY - What you go to when she doesn't believe you were at the office working late .. eg "The train was Hijacked by the French Secret Service!"

About Machinecode and Assembly Language

For those of you who, like me are novices in the micro world, the articles in previous newsletter issues (or in other less reputable sources) which talk about "Assembly Language" or "Machine code Routines" were a major challenge to understand.

With a little bit of delving, and a lot of time, I have begun to understand the basics of this confusing world. To try to explain this, I'm going right back to basics - so please bear with me:

- Any program is stored in memory as a series of Binary numbers.

- The microprocessor, once directed to the program interprets each number as an operation which it must obey.

- If we could all talk, read and think in Binary also, this article would be unnecessary.

- If we mere humans are to understand what a program is instructing the computer to do, we must have some means of interpreting the Binary Codes into logical actions.

- The so called "high-level" languages, ie BASIC, PASCAL, FORTH etc. all make this job of interpretation easy - but to provide certain fast-moving graphic effects or whatever we need something faster ...

- This is where the Assembly Language and Machinecode come in.

Machinecode - is the actual Binary code which the Microprocessor uses. For convenience purposes, we write down any Machinecodes as a hexadecimal number - it's easier to write "C5" than "11000101"

Assembly Language - is basically the list of operations which the microprocessor can execute. These operations are abbreviated into the list of instructions which appear in the following tables.

These codes alone do not enable you to program in machine code, but given one of Don's, or Garth's, or Ross's Assembly language listings, plus a little understanding of your computer, you can use the tables to convert a brief Assembly Language Listing into a set of Machinecode numbers. Once stored in memory, the microprocessor can be instructed to use the Machinecode routine from your BASIC program ... as was done in the CLOAD JUNIOR Column in the October issue.

NOTE: Variables in the table have been substituted by:

Absolute Value n OBJ CODE: &H20
 Address nn &H8405
 Relative
 Displacement e &H2E
 The Literal d &H05

OBJ CODE	SOURCE STATEMENT
BE	ADC A,(HL)
DD8E05	ADC A,(IX+d)
FD8E05	ADC A,(IY+d)
8F	ADC A,A
88	ADC A,B
89	ADC A,C
8A	ADC A,D
8B	ADC A,E
8C	ADC A,H
8D	ADC A,L
CE20	ADC A,n
ED4A	ADC HL,BC
ED5A	ADC HL,DE
ED6A	ADC HL,HL
ED7A	ADC HL,SP
86	ADD A,(HL)
DD8605	ADD A,(IX+d)
FD8605	ADD A,(IY+d)
87	ADD A,A
80	ADD A,B
81	ADD A,C
82	ADD A,D
83	ADD A,E
84	ADD A,H
85	ADD A,L
C620	ADD A,n
09	ADD HL,BC
19	ADD HL,DE
29	ADD HL,HL
39	ADD HL,SP
DD09	ADD IX,BC
DD19	ADD IX,DE
DD29	ADD IX,IX
DD39	ADD IX,SP
FD09	ADD IY,BC
FD19	ADD IY,DE
FD29	ADD IY,IY
FD39	ADD IY,SP
A6	AND (HL)
DDA605	AND (IX+d)
FDA605	AND (IY+d)
A7	AND A
A0	AND B
A1	AND C
A2	AND D
A3	AND E
A4	AND H
A5	AND L

OBJ CODE	SOURCE STATEMENT
E620	AND n
CB46	BIT 0,(HL)
DDC80546	BIT 0,(IX+d)
FDC80546	BIT 0,(IY+d)
CB47	BIT 0,A
CB40	BIT 0,B
CB41	BIT 0,C
CB42	BIT 0,D
CB43	BIT 0,E
CB44	BIT 0,H
CB45	BIT 0,L
CB4E	BIT 1,(HL)
DDC8054E	BIT 1,(IX+d)
FDC8054E	BIT 1,(IY+d)
CB4F	BIT 1,A
CB48	BIT 1,B
CB49	BIT 1,C
CB4A	BIT 1,D
CB4B	BIT 1,E
CB4C	BIT 1,H
CB4D	BIT 1,L
CB56	BIT 2,(HL)
DDC80556	BIT 2,(IX+d)
FDC80556	BIT 2,(IY+d)
CB57	BIT 2,A
CB50	BIT 2,B
CB51	BIT 2,C
CB52	BIT 2,D
CB53	BIT 2,E
CB54	BIT 2,H
CB55	BIT 2,L
CB5E	BIT 3,(HL)
DDC8055E	BIT 3,(IX+d)
FDC8055E	BIT 3,(IY+d)
CB5F	BIT 3,A
CB58	BIT 3,B
CB59	BIT 3,C
CB5A	BIT 3,D
CB5B	BIT 3,E
CB5C	BIT 3,H
CB5D	BIT 3,L
CB66	BIT 4,(HL)
DDC80566	BIT 4,(IX+d)
FDC80566	BIT 4,(IY+d)
CB67	BIT 4,A
CB60	BIT 4,B
CB61	BIT 4,C
CB62	BIT 4,D

OBJ CODE	SOURCE STATEMENT
CB63	BIT 4,E
CB64	BIT 4,H
CB65	BIT 4,L
CB6E	BIT 5,(HL)
DDC8056E	BIT 5,(IX+d)
FDC8056E	BIT 5,(IY+d)
CB6F	BIT 5,A
CB68	BIT 5,B
CB69	BIT 5,C
CB6A	BIT 5,D
CB6B	BIT 5,E
CB6C	BIT 5,H
CB6D	BIT 5,L
CB76	BIT 6,(HL)
DDC80576	BIT 6,(IX+d)
FDC80576	BIT 6,(IY+d)
CB77	BIT 6,A
CB70	BIT 6,B
CB71	BIT 6,C
CB72	BIT 6,D
CB73	BIT 6,E
CB74	BIT 6,H
CB75	BIT 6,L
CB7E	BIT 7,(HL)
DDC8057E	BIT 7,(IX+d)
FDC8057E	BIT 7,(IY+d)
CB7F	BIT 7,A
CB78	BIT 7,B
CB79	BIT 7,C
CB7A	BIT 7,D
CB7B	BIT 7,E
CB7C	BIT 7,H
CB7D	BIT 7,L
DC8405	CALL C,n
FC8405	CALL M,nn
D48405	CALL NC,nn
CD8405	CALL NZ,nn
F48405	CALL P,nn
EC8405	CALL PE,nn
E48405	CALL PO,nn
CC8405	CALL Z,nn
CD8405	CALL nn
3F	CCF
BE	CP (HL)
DD8E05	CP (IX+d)
FD8E05	CP (IY+d)
BF	CP A
B8	CP B
B9	CP C
BA	CP D
BB	CP E
BC	CP H
BD	CP L
FE20	CP n
EDAS	CPD
EDB9	CPDR

OBJ CODE	SOURCE STATEMENT	OBJ CODE	SOURCE STATEMENT	OBJ CODE	SOURCE STATEMENT	OBJ CODE	SOURCE STATEMENT
C89F	RES 3 A	ED4D	RETI	9L	SBC	DOC805E6	SET 4(IIX+di)
C89B	RES 3 B	C816	RL	DOC805E	SBC A(IIX+di)	DOC805E6	SET 4(IIX+di)
C899	RES 3 C	DOC80516	RL	DOC805E	SBC A(IIX+di)	C8E7	SET 4 A
C89A	RES 3 D	DOC80516	RL	9F	SBC A A	C8E0	SET 4 B
C898	RES 3 E	C817	RL	98	SBC A B	C8E1	SET 4 C
C89C	RES 3 H	C817	RL	99	SBC A C	C8E2	SET 4 D
C89D	RES 3 L	C811	RL	9A	SBC A D	C8E3	SET 4 E
C89A6	RES 4(IHL)	C812	RL	9B	SBC A E	C8E4	SET 4 H
DOC805A6	RES 4(IIX+di)	C813	RL	9C	SBC A H	C8E5	SET 4 L
DOC805A6	RES 4(IIX+di)	C814	RL	9D	SBC A L	C8E6	SET 5(IHL)
C8A7	RES 4 A	C815	RL	ED42	SBC HLBC	DOC805EE	SET 5(IIX+di)
C8A0	RES 4 B	17	RLA	ED52	SBC HLDE	DOC805EE	SET 5(IIX+di)
C8A1	RES 4 C	C806	RLC	ED72	SBC HLSP	C8E8	SET 5 A
C8A2	RES 4 D	DOC80508	RLC	71	SBC	C8E9	SET 5 B
C8A3	RES 4 E	DOC80506	RLC	71	SCF	C8EA	SET 5 C
C8A4	RES 4 F	DOC80506	RLC	71	SCF	C8EB	SET 5 D
C8A5	RES 4 H	C807	RLC	DOC80506	SET 0(IHL)	C8EC	SET 5 E
DOC805A6	RES 5(IHL)	C800	RLC	DOC80506	SET 0(IIX+di)	C8ED	SET 5 F
DOC805A6	RES 5(IIX+di)	C801	RLC	DOC80506	SET 0(IIX+di)	C8EE	SET 5 G
DOC805A6	RES 5(IIX+di)	C802	RLC	C8C7	SET 0 A	C8EF	SET 5 H
C8AF	RES 5 A	C803	RLC	C8C7	SET 0 B	C8F6	SET 6(IHL)
C8AB	RES 5 B	C804	RLC	C8C7	SET 0 C	DOC805F6	SET 6(IIX+di)
C8A9	RES 5 C	C805	RLC	C8C3	SET 0 D	DOC805F6	SET 6(IIX+di)
C8AA	RES 5 D	07	RLCA	C8C4	SET 0 E	C8F7	SET 6 A
C8AB	RES 5 E	ED5F	RLD	C8C5	SET 0 H	C8F0	SET 6 B
C8AC	RES 5 H	C81E	RR	C8C6	SET 0 L	C8F1	SET 6 C
C8AD	RES 5 L	DOC8051E	RR	DOC805CE	SET 1(IHL)	C8F2	SET 6 D
C8B6	RES 6(IHL)	DOC8051E	RR	C8CF	SET 1(IIX+di)	C8F3	SET 6 E
DOC805B6	RES 6(IIX+di)	C81F	RR	DOC805CE	SET 1(IIX+di)	C8F4	SET 6 F
DOC805B6	RES 6 A	C818	RR	C8C8	SET 1 A	C8F5	SET 6 L
C8B0	RES 6 B	C819	RR	C8C9	SET 1 B	C8FE	SET 7(IHL)
C8B1	RES 6 C	C81A	RR	C8CA	SET 1 C	DOC805FE	SET 7(IIX+di)
C8B2	RES 6 D	C81C	RR	C8CB	SET 1 D	DOC805FE	SET 7(IIX+di)
C8B3	RES 6 E	C81D	RR	C8CC	SET 1 E	C8FF	SET 7 A
C8B4	RES 6 H	C81D	RR	C8CD	SET 1 H	C8F8	SET 7 B
C8B5	RES 6 L	1F	RR	C8CE	SET 1 L	C8F9	SET 7 C
C8BE	RES 6 L	C80E	RR	DOC80506	SET 2(IHL)	C8FA	SET 7 D
DOC805B6	RES 7(IIX+di)	DOC8050E	RR	DOC80506	SET 2(IIX+di)	C8FB	SET 7 E
DOC805B6	RES 7 A	DOC8050E	RR	DOC80506	SET 2(IIX+di)	C8FC	SET 7 F
C8BF	RES 7 A	C80F	RR	C8D0	SET 2 A	C8FD	SET 7 L
C8B8	RES 7 B	C808	RR	C8D1	SET 2 B	C8FE	SET 7 L
C8B9	RES 7 C	C809	RR	C8D2	SET 2 C	DOC80526	SLA IHL
C8BA	RES 7 D	C80A	RR	C8D3	SET 2 D	DOC80526	SLA IIX+di
C8BB	RES 7 E	C80B	RR	C8D4	SET 2 E	C8D7	SLA A
C8BC	RES 7 F	C80C	RR	C8D5	SET 2 H	C8D0	SLA B
C8BD	RES 7 H	C80D	RR	C8D6	SET 2 L	C8D1	SLA C
C8BD	RES 7 L	ED67	RRD	C8D8	SET 3 B	C8D2	SLA D
C9	RET C	C7	RRD	C8DE	SET 3(IHL)	C8D3	SLA E
D6	RET M	CF	RRD	DOC8050E	SET 3(IIX+di)	C8D4	SLA F
D8	RET M	CF	RRD	DOC8050E	SET 3(IIX+di)	C8D5	SLA G
D0	RET NC	D7	RRD	C8D9	SET 3 A	C8D6	SLA H
C0	RET NZ	D7	RRD	C8DA	SET 3 C	DOC8052E	SRA IIX+di
F0	RET P	E7	RRD	C8DB	SET 3 D	DOC8052E	SRA IIX+di
E8	RET PE	F7	RRD	C8DC	SET 3 E	C8D2F	SRA A
E0	RET PO	FF	RRD	C8DD	SET 3 H	C8D9	SRA B
C8	RET Z	DE20	RRD	C8DE	SET 3 L	C8D9	SRA C
				C8E6	SET 4(IHL)	C8D9A	SRA D

OBJ CODE	SOURCE STATEMENT
C82B	SRA E
C82C	SRA H
C82D	SRA L
C83E	IHL
DOC8051E	SRL IIX+di
DOC8051E	SRL IIX+di
C83F	SRL A
C839	SRL B
C83A	SRL C
C83J	SRL D
C83K	SRL E
C83L	SRL H
C83D	SRL L
96	SUB IHL
DD8005	SUB IIX+di
FD8005	SUB IIX+di
97	SUB A
90	SUB B
91	SUB C
92	SUB D
93	SUB E
94	SUB H
95	SUB L
DD800	SUB n
AE	XOR IHL
DDA005	XOR IIX+di
FDA005	XOR IIX+di
AF	XOR A
AB	XOR B
A9	XOR C
AA	XOR D
AB	XOR E
AC	XOR H
AD	XOR L
EE20	XOR n

Action Computers

Software Descriptions For SpectraVideo / MSX

SVI1_V1.0

Identity Quest (Text Adventure)

You are an adventurer searching for treasures and at the same time trying to stay alive, as monsters and unseen perils attempt to halt your progress. You have 6 attributes randomly set up at the start which control your destiny. Every game is different, and the ability to save a game and restart later is available. Loads of fun for all, this game is currently cassette based only.

SVI2_V1.0

Graph Plot

This is an educational numerical analysis program which allows plotting of 2d graphs with asymptotes. As well as this you have several numerical analysis routines such as equation solving, integration, differentiation and solving of linear systems. Graphs and result screens can be dumped to the printer.

SVI3_V1.0

Super Art

An artist program which has many new features such as copying parts of the picture elsewhere, extensive on-line help, screen plots, saving screens in RAM and much more. Available for SVI328 only at this stage.

SVI4_V1.0

Graph Presenter

Graph Presenter allows you to enter data such as monthly sales volumes over a 1 or 2 year period and then graphically present the data in a number of ways. Histograms, line graphs, pie charts and bar graphs are available, along with the ability to send plots to the printer.

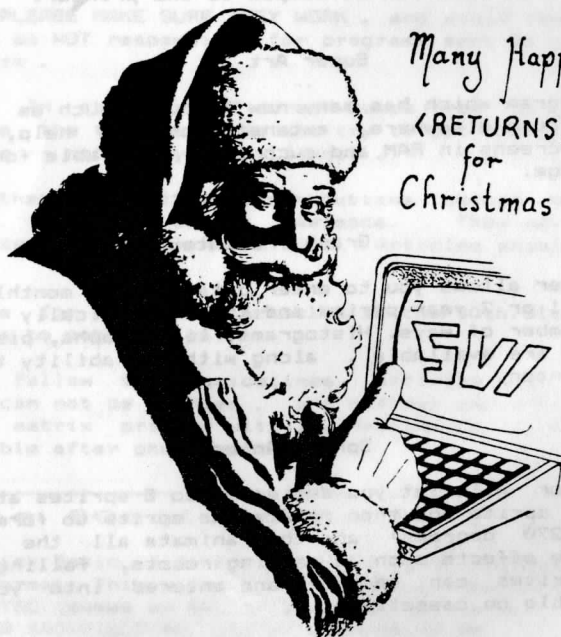
SVI5_V1.0

Sprite Animator

Sprite Animator will let you design up to 8 sprites at once. You can design a sprite and then rotate the sprite to form a new sprite (90,180,270 degrees) and then animate all the sprites created to create effects such as walking robots, falling rocks and so on. Sprites can be saved and entered into your own software. Available on cassette only.

This is a CPM program designed for use with TURBO PASCAL. With this program you can design an application screen for your own programs. SCREENMK will generate TURBO PASCAL procedures which can be incorporated into your own program to recreate the screen. This program takes all the tedium out of laying out screens, and also allows you to define positions on the screen where your applications program will read variables from, name the variables, and write the TURBO code to access those variable fields. Can also be made available for any other system running TURBO PASCAL.

Action Computers



Many Happy
<RETURNS>
for
Christmas

from
Your
Editors

Classifieds:

Some Club members have been inquiring about Disc Drives and Controllers. Anyone with such hardware for sale could contact the editors for details.